LEC. 2

THERMODYNAMICS 1

CHAPTER 2

ENERGY, ENERGY TRANSFER & GENERAL ENERGY ANALYSIS

ENGR 208 · Study Note · Concepts + Solved Problems

نوت مرتّب: الفكرة، القانون، ومسائل محلولة خطوة خطوة

The First Law of Thermodynamics

conservation of energy principle [Energy balance]

* Energy can't be created or destroyed, but it can change from one form to another

* The energy gained by a system = the energy lost by the surroundings

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(Total energy entering the system) − (Total energy leaving the system) = (Change in the total energy of the system)

$$E_{in} - E_{out} = \Delta E_{system}$$

$$\Delta E_{system} = E_{final} - E_{initial} = E_2 - E_1$$

حساب بنكي: اللي يدخل − اللي يطلع = التغيّر في الرصيد

How energy crosses the boundary

Closed system

no mass crosses the boundary

→ **Heat** Q

→ **Work** W

Open system (control volume)

mass crosses the boundary

→ **Heat** Q

→ **Work** W

→ **Mass flow** E_{mass}

first question in any problem: does mass cross the boundary? فيه مادة تعبر الحدود؟

Forms of Energy

* Energy exists in several forms: thermal, mechanical, kinetic, potential, electric, magnetic, chemical & nuclear

* The sum of all forms is the **total energy [E]** (kJ) per unit mass $e = \frac{E}{m}$ (kJ/kg)

Energy

macroscopic

with respect to an **outside reference frame**

e.g. kinetic energy, potential energy

microscopic

related to the **molecular structure** and the degree of molecular activity

* their sum = internal energy [U]

Kinetic energy

$KE = \frac{1}{2} m V^2$ (J)

$KE = \frac{m V^2}{2000}$ (kJ)

$ke = \frac{V^2}{2000}$ (kJ/kg)

Potential energy

$PE = m g z$ (J)

$PE = \frac{m g z}{1000}$ (kJ)

$pe = \frac{g z}{1000}$ (kJ/kg)

Total energy

$$E = U + KE + PE = U + \frac{mV^2}{2000} + \frac{mgz}{1000} \quad (kJ)$$

per unit mass $e = u + \frac{V^2}{2000} + \frac{gz}{1000}$ (kJ/kg)

stationary system → $\Delta KE = 0$, $\Delta PE = 0$ → $\Delta E = \Delta U$ النظام ثابت: التغير في الطاقة = التغير في U

Internal Energy

1. Sensible energy → kinetic energy of the molecules (تزيد مع الحرارة)
2. Latent energy → the **phase** of the system (سائل ↔ بخار)

   * sensible + latent = thermal energy
3. Chemical energy → atomic bonds in a molecule
4. Nuclear energy → strong bonds within the nucleus of the atom itself

Types of Energy

1. Static energy → total energy **stored** in a system
2. Dynamic energy → energy **interacting** with the system (heat & work)
3. Thermal energy → sensible + latent forms of internal energy

   !! thermal energy ≠ heat (heat is energy in transit)
4. Mechanical energy → energy that can be converted to mechanical work completely and directly by an ideal device (turbine, pump, fan)

سؤال نظري متكرر: عرّف كل نوع بجملة وحدة. احفظ الكلمة المفتاحية: molecules · phase · bonds · nucleus

Mass flow rate

$$\dot{m} = \frac{m}{t} = \rho \dot{V} = \rho A V \quad (kg/s)$$

$A = \pi D^2/4$ for a pipe

Volume flow rate

$$\dot{V} = \frac{Vol}{t} = A V \quad (m^3/s)$$

$$1\ L = \frac{1}{1000}\ m^3$$

Energy flow rate $\dot{E}_{mass} = \dot{m}\ \theta \quad (kW)$

Flow work (flow energy)

* the work needed to **push** mass into or out of the control volume الطاقة اللازمة لدفع المائع

$$W_{flow} = P \cdot Vol \ (kJ) \qquad w_{flow} = P\ v \ (kJ/kg)$$

Enthalpy

* internal energy + flow work, used mostly in open systems

$$H = U + P \cdot Vol \ (kJ) \qquad h = u + P\ v \ (kJ/kg)$$

energy per kg of flowing mass: $\theta = h + \frac{V^2}{2000} + \frac{g\ z}{1000} \quad (kJ/kg)$

closed system → use **u** | mass crossing → use **h**

Water flows through a pipe of diameter **5 cm** at an average velocity of **3 m/s**. Determine the volume flow rate (in L/s) and the mass flow rate.

$A = \frac{\pi}{4}(0.05)^2 = 1.9635\times10^{-3}\ m^2$ $\quad D = 5/100 = 0.05\ m$

$\dot{V} = A\ V = (1.9635\times10^{-3})(3) = 5.89\times10^{-3}\ m^3/s =$ **5.89 L/s**

$\dot{m} = \rho\ \dot{V} = (1000)(5.89\times10^{-3}) =$ **5.89 kg/s**

Steam enters a turbine at a rate of **2 kg/s** with an enthalpy of **3000 kJ/kg** and a velocity of **50 m/s**. Neglecting potential energy, find the rate of energy carried by the steam.

$$\theta = h + \frac{V^2}{2000} + \cancel{\frac{gz}{1000}} = 3000 + \frac{(50)^2}{2000} = 3001.25 \text{ kJ/kg}$$

$$\dot{E}_{mass} = \dot{m}\,\theta = (2)(3001.25) = \boxed{6002.5 \text{ kW}}$$

لاحظ: حد السرعة (1.25) صغير جداً مقارنة بـ h، بس لا تهمله إلا إذا قال السؤال

Energy Transfer by Heat

energy transferred between two systems (or a system and its surroundings) as a result of a temperature difference only

* *heat is energy **in transition*** · * *adiabatic (insulated) ⇒ no heat transfer, $Q = 0$*

Forms of heat

heat transfer	Q	(kJ)
per unit mass	$q = \frac{Q}{m}$	(kJ/kg)
rate of heat transfer	$\dot{Q} = \frac{Q}{t}$	(kJ/s ≡ kW)

A pot containing **2 kg** of water receives **180 kJ** of heat in **2 minutes**. Determine the heat transfer per unit mass and the rate of heat transfer.

$$q = \frac{Q}{m} = \frac{180}{2} = \boxed{90 \text{ kJ/kg}}$$

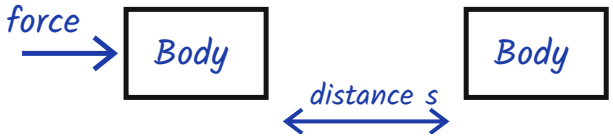
$$\dot{Q} = \frac{Q}{t} = \frac{180}{2 \times 60} = \boxed{1.5 \text{ kW}}$$

Mechanisms of heat transfer

mechanism		between
conduction	التوصيل	solid ⇄ solid (direct contact)
convection	الحمل	solid ⇄ moving liquid or gas
radiation	الإشعاع	electromagnetic waves (e.g. the sun), no medium needed

Energy Transfer by Work

energy transfer associated with a **force acting through a distance** (raise a weight, turn a shaft, ...)



Work = Force × Distance

Q: When do we have work?

A: two conditions together: 1 a force acts on the boundary 2 the boundary moves

قوة على الحدود + الحدود تتحرك. إذا وحدة ناقصة ما فيه شغل

Forms of work

work	W	(kJ)
work per unit mass	$w = \frac{W}{m}$	(kJ/kg)
rate of work (power)	$\dot{W} = \frac{W}{t}$	(kJ/s ≡ kW)

Heat & work: what they are (and aren't)

heat and work are **energy transfer mechanisms**:

1. boundary phenomena
2. not held by systems (النظام ما يملكها)
3. associated with a process
4. path functions

Path functions & point functions

	path functions	**point functions**
depend on	states **and** path	states only
examples	heat & work	properties: P, V, T, ...
differential	inexact: δQ , δW	exact: dP , dV , dT
integral	$\int_1^2 \delta W = W_{12}$ (not $W_2 - W_1$)	$\int_1^2 dT = T_2 - T_1 = \Delta T$

Directions of heat & work

course notes (in / out)

write every value **positive** and put the direction in the subscript:

Q_{in} , Q_{out} , W_{in} , W_{out}

$Q_{net,in} = Q_{in} - Q_{out}$

$W_{net,out} = W_{out} - W_{in}$

sign form (same result)

$Q_{in} \rightarrow +$ $Q_{out} \rightarrow -$

$W_{out} \rightarrow +$ $W_{in} \rightarrow -$

نفس الجواب بالطريقتين، بس لا تخلطهم في مسألة وحدة

Types of work

electrical work	shaft work	spring work	work to raise or accelerate a body

1 Electrical work

V → voltage (V)

I → current (A)

t → time (s)

R → resistance (Ω) $V = I R$

$$W_e = \frac{V I t}{1000} \quad (kJ)$$

$$\dot{W}_e = \frac{V I}{1000} \quad (kW)$$

convert first: mA ÷ 1000 → A · minutes × 60 → s · hours × 3600 → s

An electric kettle operates at **220 V** and draws a current of **8 A**. Determine the electrical power and the energy supplied to the water if it runs for **3 minutes**.

$$\dot{W}_e = \frac{V I}{1000} = \frac{(220)(8)}{1000} = \boxed{1.76 \text{ kW}}$$

$$W_e = \frac{V I t}{1000} = \frac{(220)(8)(3 \times 60)}{1000} = \boxed{316.8 \text{ kJ}}$$

A resistance heater draws **500 mA** from a **120 V** source for **1 hour**. How much work is transferred?

$$I = \frac{500}{1000} = 0.5 \text{ A} \qquad t = 1 \times 3600 = 3600 \text{ s}$$

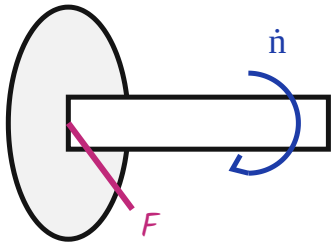
$$W_e = \frac{(120)(0.5)(3600)}{1000} = \boxed{216 \text{ kJ}}$$

2 Shaft work

$$W_{sh} = \frac{2\pi\, n\, \tau}{1000} \quad (kJ)$$

τ → torque (N·m) $\quad \tau = F\, r$

n → number of revolutions



shaft power

$$\dot{W}_{sh} = \frac{2\pi\, \dot{n}\, \tau}{1000} \quad (kJ/s \equiv kW) \qquad \dot{n} = \frac{n}{t} = \frac{\text{number of revolutions}}{\text{sec}}$$

rpm → rev/s : ÷ 60 أكثر غلطة في الشافت: نسيان القسمة على 60

A motor delivers a torque of **150 N·m** to a shaft rotating at **3000 rpm**. Determine (a) the shaft power, and (b) the work done in **2 minutes**.

(a) $\dot{n} = 3000 \frac{rev}{min} = \frac{3000}{60} = 50 \frac{rev}{s}$

$$\dot{W}_{sh} = \frac{2\pi\,(50)(150)}{1000} = \boxed{47.12\ kW}$$

(b) $n = 3000 \times 2 = 6000$ rev

$$W_{sh} = \frac{2\pi\,(6000)(150)}{1000} = \boxed{5654.9\ kJ}$$

(check: 47.12 × 120 s ✓)

3 Spring work

$$W_{spring} = \frac{\frac{1}{2}\, k\, (x_2^2 - x_1^2)}{1000} \quad (kJ)$$

k → spring constant (N/m) · x measured from the **rest position (free length)**

* at rest position → x_1 = zero · * initial force F_1 given → $x_1 = F_1 / k$

units: kN/m × 1000 → N/m · kN/cm × 1000 × 100 → N/m · cm ÷ 100 → m

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A linear spring with a spring constant of **50 kN/m** is compressed from **5 cm** to **15 cm** (measured from its rest position). Determine the work required.

$k = 50 \times 1000 = 50\,000$ N/m $\quad x_1 = \frac{5}{100} = 0.05$ m $\quad x_2 = \frac{15}{100} = 0.15$ m

$$W = \frac{\frac{1}{2}\,(50\,000)(0.15^2 - 0.05^2)}{1000} = \boxed{0.5 \text{ kJ}}$$

A spring with **k = 3 kN/cm** already has an initial force of **0.6 kN** acting on it. Determine the work required to compress it another **1 cm**.

$k = 3 \times 1000 \times 100 = 300\,000$ N/m $\quad F_1 = 0.6 \times 1000 = 600$ N

$F_1 = k\, x_1 \Rightarrow 600 = (300\,000)\, x_1 \Rightarrow x_1 = 2 \times 10^{-3}$ m

$x_2 = x_1 + \frac{1}{100} = 0.002 + 0.01 = 0.012$ m

$$W = \frac{\frac{1}{2}\,(300\,000)(0.012^2 - 0.002^2)}{1000} = \boxed{0.021 \text{ kJ}} = 21 \text{ J}$$

انتبه: x_2 هو 1 cm، الإزاحة الأولى + الإضافية

4 Work done to raise or accelerate a body

acceleration — عند تغيّر السرعة

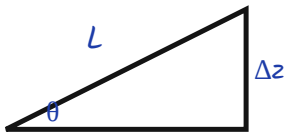
$$W_a = \Delta KE = \frac{m\,(V_2^2 - V_1^2)}{2000}$$

raising — عند تغيّر الارتفاع

$$W_g = \Delta PE = \frac{m\,g\,(z_2 - z_1)}{1000}$$

$$W = W_a + W_g \quad (kJ)$$

$$\dot{W} = \frac{W}{t} \quad (kW)$$



$$z_2 - z_1 = L \sin\theta$$

km/h → m/s : × 1000 ÷ 3600

constant speed ⇒ $W_a = 0$ · speed decreases ⇒ W_a is negative

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A **1000 kg** car accelerates from rest to **72 km/h** in **10 s** while climbing a hill with a vertical rise of **20 m**. Determine the work and the average power required.

$$V_1 = 0 \qquad V_2 = 72\ \frac{km}{h} = \frac{72 \times 1000}{3600} = 20\ m/s \qquad z_2 - z_1 = 20\ m$$

$$W = \frac{m(V_2^2 - V_1^2)}{2000} + \frac{m\,g\,(z_2 - z_1)}{1000} = \frac{1000\,(20^2)}{2000} + \frac{1000\,(9.81)(20)}{1000}$$

$$= 200 + 196.2 = \boxed{396.2\ kJ}$$

$$\dot{W} = \frac{W}{t} = \frac{396.2}{10} = \boxed{39.62\ kW}$$

An **800 kg** elevator rises **30 m** at a constant speed in **15 s**. Determine the minimum power required.

$V = const \Rightarrow W_a = zero$ (سرعة ثابتة)

$$W = W_g = \frac{m\,g\,(z_2 - z_1)}{1000} = \frac{800\,(9.81)(30)}{1000} = 235.44\ kJ$$

$$\dot{W} = \frac{235.44}{15} = \boxed{15.70\ kW}$$

Mechanical Energy E_{mech}

turbine · pump · fan

energy that can be converted to mechanical work completely

normal form

$$E_{mech} = P \cdot Vol + \frac{mV^2}{2000} + \frac{mgz}{1000} \quad (kJ)$$

per unit mass

$$e_{mech} = \frac{P}{\rho} + \frac{V^2}{2000} + \frac{gz}{1000} \quad (kJ/kg)$$

$$v = \frac{Vol}{m} = \frac{1}{\rho}$$

rate form (power generation)

$$\Delta \dot{E}_{mech} = \dot{m}\left[\frac{P_2 - P_1}{\rho} + \frac{V_2^2 - V_1^2}{2000} + \frac{g(z_2 - z_1)}{1000}\right] \quad (kW)$$

P in kPa $\Rightarrow \frac{\Delta P}{\rho}$ is already in kJ/kg · شيل كل حد ما تغيّر

Water is released from a dam through a turbine from a height of **60 m** at a rate of **3 m³/s**. Determine the power generation potential, and the energy produced in **5 hours** (kWh).

$\dot{m} = \rho \dot{V} = (1000)(3) = 3000$ kg/s

$\Delta \dot{E}_{mech} = \dot{m}\left[\cancel{\frac{\Delta P}{\rho}} + \cancel{\frac{\Delta V^2}{2000}} + \frac{g\,\Delta z}{1000}\right] = 3000\,\frac{(9.81)(60)}{1000} = \boxed{1765.8 \text{ kW}}$

energy $= 1765.8 \times 5 = \boxed{8829 \text{ kWh}}$

A wind turbine has blades of **10 m** diameter. Wind blows at **12 m/s** with a density of **1.2 kg/m³**. Find the mechanical energy per unit mass and the power generation potential.

$A = \frac{\pi}{4}(10)^2 = 78.54$ m² $\quad \dot{m} = \rho A V = (1.2)(78.54)(12) = 1130.97$ kg/s

$e_{mech} = \cancel{\frac{P}{\rho}} + \frac{V^2}{2000} + \cancel{\frac{gz}{1000}} = \frac{(12)^2}{2000} = \boxed{0.072 \text{ kJ/kg}}$

$\Delta \dot{E}_{mech} = \dot{m}\, e_{mech} = (1130.97)(0.072) = \boxed{81.43 \text{ kW}}$

A fan in a **0.5 m × 0.5 m** duct accelerates air from rest to an average velocity of **10 m/s**. Taking ρ = **1.2 kg/m³**, estimate the minimum electric power of the fan motor.

$A = 0.5 \times 0.5 = 0.25$ m² $\quad \dot{m} = \rho A V = (1.2)(0.25)(10) = 3$ kg/s

$\Delta \dot{E}_{mech} = \dot{m}\,\frac{V_2^2 - V_1^2}{2000} = 3\,\frac{(10)^2 - 0}{2000} = \boxed{0.15 \text{ kW}} = 150 \text{ W}$

The energy balance: all forms

general:

$$(Q_{in} - Q_{out}) + (W_{in} - W_{out}) + (E_{mass,in} - E_{mass,out}) = \Delta U + \Delta KE + \Delta PE$$

$$Q_{net,in} - W_{net,out} = \Sigma\, m_e\theta_e - \Sigma\, m_i\theta_i + (m_2e_2 - m_1e_1)_{cv}$$

closed system

stationary → $\Delta KE = 0$, $\Delta PE = 0$

$E_{mass} = 0$

$$Q_{net,in} - W_{net,out} = \Delta U$$

cycle

returns to its initial state

$\Delta U = 0$ (U depends on the states only)

$$Q_{net,in} = W_{net,out}$$

steady flow

nothing changes inside → $\Delta E_{cv} = 0$

$$\dot{E}_{in} = \dot{E}_{out}$$

adiabatic

insulated

$$Q = 0$$

How to start any problem

1. closed or open? فيه كتلة تعبر؟
2. sketch: an arrow for every Q, W, mass — in or out
3. name the process: steady? cycle? adiabatic? stationary?
4. write $E_{in} - E_{out} = \Delta E_{sys}$ and cross out the zeros
5. plug in · 6. check units (kJ · kJ/kg · kW)

Example 2-1 Air flows into an open system and carries energy at the rate of **300 kW**. As it flows through, it receives **600 kW** of work and loses **100 kW** of energy by heat transfer to the surroundings. If the system experiences no energy change, how much energy does the air carry as it leaves the system?

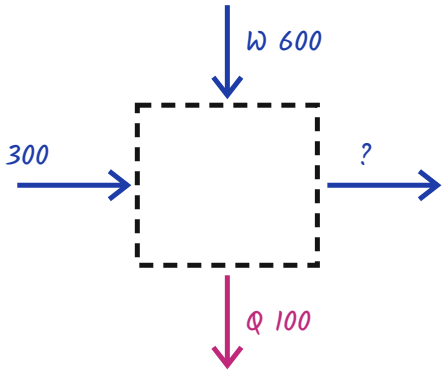
open system , steady $\rightarrow \Delta E_{sys} = 0$

$\dot{E}_{in} - \dot{E}_{out} = \cancel{\Delta \dot{E}_{sys}} \rightarrow 0$

$\dot{E}_{mass,in} + \dot{W}_{in} = \dot{Q}_{out} + \dot{E}_{mass,out}$

$300 + 600 = 100 + \dot{E}_{mass,out}$

$\boxed{\dot{E}_{mass,out} = 800 \text{ kW}}$



Example 2-2 A steam power plant operates on a cycle. For each kilogram of steam, the cycle receives **2000 kJ** of heat in the boiler, rejects **1500 kJ** in the condenser, and receives **5 kJ** of work in the pump. Determine the work done by the steam in the turbine (kJ/kg).

cycle $\Rightarrow \Delta U = 0 \Rightarrow$ $q_{net,in} = w_{net,out}$

$q_{net,in} = q_{boiler} - q_{condenser} = 2000 - 1500 = 500 \text{ kJ/kg}$

$w_{net,out} = w_{turbine} - w_{pump} = w_{turbine} - 5$

$500 = w_{turbine} - 5 \Rightarrow$ $\boxed{w_{turbine} = 505 \text{ kJ/kg}}$

الدورة: صافي الحرارة = صافي الشغل. التوربين يطلع شغل، والمضخة تاخذ شغل

A rigid tank contains a gas. During a process, **15 kJ** of heat is transferred to the gas, **3 kJ** of heat is lost to the surroundings, and a paddle wheel does **4 kJ** of work on the gas. Determine the change in the internal energy of the gas.

$Q_{in} = 15\ kJ \quad Q_{out} = 3\ kJ \quad W_{in} = 4\ kJ$

closed system:

paddle wheel, electric heater, shaft ⇒ W_{in}

1st law $\quad Q_{net,in} - W_{net,out} = \Delta U$

$[15 - 3] - [0 - 4] = \Delta U$

$\boxed{\Delta U = 16\ kJ}$

A piston–cylinder device contains a gas with an initial internal energy of **40 kJ**. The gas receives **50 kJ** of heat, loses **10 kJ** through the walls, and does **12 kJ** of boundary work as the piston rises. Find the final internal energy.

$Q_{in} = 50 \quad Q_{out} = 10 \quad W_{out} = 12 \quad U_1 = 40\ kJ$

$[50 - 10] - [12 - 0] = U_2 - U_1$

$28 = U_2 - 40 \Rightarrow \boxed{U_2 = 68\ kJ}$

W out ⇒ الغاز سوّى شغل ⇒ المكبس طلع

A closed room contains a **2 kW** electric heater and a **300 W** fan, both running. Heat is lost through a window at a rate of **1.5 kW**. Determine the rate of change of the energy of the room.

$\dot{W}_{in} = 2 + \frac{300}{1000} = 2.3\ kW \quad \dot{Q}_{out} = 1.5\ kW$

$\dot{Q}_{net,in} - \dot{W}_{net,out} = \Delta \dot{U} \Rightarrow [0 - 1.5] - [0 - 2.3] = \Delta \dot{U}$

$\boxed{\Delta \dot{U} = 0.8\ kW}$ (the room is warming up)

A computer lab holds **30 students**, each dissipating heat at **400 kJ/h**, and **20 computers** rated at **150 W** each. Heat enters through the walls at **10 800 kJ/h**. The room is kept at a constant temperature using air-conditioning units of **4 kW** each. How many units are needed?

$\dot{Q}_{people} = 30 \times \frac{400}{3600} = 3.33\ kW\ (in)$ $\quad \dot{W}_{pc} = 20 \times \frac{150}{1000} = 3\ kW\ (in)$

$\dot{Q}_{walls} = \frac{10\ 800}{3600} = 3\ kW\ (in)$ $\quad$ constant $T \Rightarrow \Delta\dot{U} = 0$

$[3.33 + 3 - \dot{Q}_{AC}] - [0 - 3] = 0 \Rightarrow \dot{Q}_{AC} = 9.33\ kW$

$units = \frac{9.33}{4} = 2.33 \Rightarrow$ **3 units** (قرّب دائماً لفوق)

A heat engine receives **1200 kJ** of heat and rejects **750 kJ** of heat per cycle. Determine the net work output per cycle.

cycle $\Rightarrow Q_{net,in} = W_{net,out} \Rightarrow W_{net,out} = 1200 - 750 =$ **450 kJ**

A system goes from state A to state B along path **1**, receiving **80 kJ** of heat and doing **30 kJ** of work. It returns from B to A along path **2**, rejecting **25 kJ** of heat. Find the work in path 2 and say whether it is done by or on the system.

cycle: $Q_1 + Q_2 = W_1 + W_2$ (sign form)

$80 + (-25) = 30 + W_2$

$W_2 = +25\ kJ$ → done **by** the system

A — path 1 → / ← path 2 — B

إذا طلع الشغل موجب في طريقة الإشارات ⇒ النظام هو اللي سوّى الشغل

2-8 Energy and Environment

المطلوب: شنو هو، شنو سببه، وشلون نقلّله

topic	what it is / cause	how to reduce it
Ozone (ground level)	forms when NOx and hydrocarbons (VOCs) from vehicles and industry react in **sunlight**; irritates the lungs	cut NOx and VOC emissions: cleaner engines, catalytic converters
Smog	brown haze over cities, mainly ground-level ozone plus CO, particulates and VOCs from burning fuel	cleaner fuels, efficient combustion, public transport
Acid rain	SO_2 and NOx from burning fossil fuels (especially high-sulfur coal) react with water vapor to form sulfuric and nitric acids	low-sulfur fuels, scrubbers on power plants
Greenhouse effect	CO_2, water vapor and methane let sunlight in but trap the infrared radiation the earth emits, keeping it warm	it is natural; the problem is when the gases increase
Global warming	rise in the earth's average temperature as CO_2 from burning fossil fuels builds up	use energy efficiently, switch to renewables
Global climate change	the wider result of warming: changed rainfall, stronger storms, melting ice, rising sea level	same as above: reduce greenhouse gas emissions

key link: burning fossil fuels is the common cause of all six. السبب المشترك: حرق الوقود الأحفوري

Summary: spot the family, start right

clue in the question	family	start with
moving water / air, "power potential"	mechanical energy	$\dot{m} = \rho AV$ or $\rho\dot{V}$, then $\dot{m}[\Delta P/\rho + \Delta V^2/2000 + g\Delta z/1000]$
heater · shaft · spring	one device	$VIt/1000 \cdot 2\pi\dot{n}\tau/1000 \cdot \frac{1}{2}k(x_2^2 - x_1^2)/1000$
car · elevator · escalator	raise / accelerate	$\Delta z = L\sin\theta$, $W = \Delta KE + \Delta PE$, $\dot{W} = W/t$
heat in & out, asks ΔU / U_2	closed first law	$Q_{net,in} - W_{net,out} = \Delta U$
constant temperature room	closed, $\Delta\dot{U} = 0$	sum everything in = what the AC removes
returns to its initial state	cycle	$Q_{net,in} = W_{net,out}$
mass in & out, nothing changes	steady open	$\dot{E}_{in} = \dot{E}_{out}$, $\theta = h + V^2/2000 + gz/1000$

Traps !!

1. $V^2/2$ gives J/kg → divide by **1000** (that's why we write $V^2/2000$)
2. rpm ÷ 60 · km/h ÷ 3.6 · mA ÷ 1000 · cm ÷ 100 · kJ/h ÷ 3600 → kW
3. kJ (total) · kJ/kg (per mass) · kW (per time): never mix in one equation
4. work is a path function → W_{12} , never $W_2 - W_1$
5. mass crosses the boundary → use h , not u
6. heat ≠ thermal energy · a number of units → always round **up**

ارسم ← حدّد العائلة ← شيل الأصفار ← عوّض ← تأكد من الوحدات